1. Identification

Product identifier used on the label

Styropor® P 426 Ccycled®

Recommended use of the chemical and restriction on use

Recommended use*: Expanding-agent containing plastic for the production of foam plastics
Recommended use*: for industrial processing only; Expanding-agent containing plastic for the production of foam plastics
Unsuitable for use: Not intended for sale to or use by the general public.

* The “Recommended use” identified for this product is provided solely to comply with a Federal requirement and is not part of the seller's published specification. The terms of this Safety Data Sheet (SDS) do not create or infer any warranty, express or implied, including by incorporation into or reference in the seller's sales agreement.

Details of the supplier of the safety data sheet

Company:
BASF CORPORATION
100 Park Avenue
Florham Park, NJ 07932, USA

Telephone: +1 973 245-6000

Emergency telephone number
24 Hour Emergency Response Information
CHEMTREC: 1-800-424-9300
BASF HOTLINE: 1-800-832-HELP (4357)

Other means of identification
Synonyms: Expandable Polystyrene

2. Hazards Identification

According to Regulation 2012 OSHA Hazard Communication Standard; 29 CFR Part 1910.1200

Classification of the product

No need for classification according to GHS criteria for this product.

Label elements

The product does not require a hazard warning label in accordance with GHS criteria.

Hazards not otherwise classified

May cause some eye irritation which should cease after removal of the product. In use may form flammable/explosive vapour-air mixture.

3. Composition / Information on Ingredients

According to Regulation 2012 OSHA Hazard Communication Standard; 29 CFR Part 1910.1200

Pentane
CAS Number: 109-66-0
Content (W/W): < 5.3%
Synonym: n-Pentan

isopentane
CAS Number: 78-78-4
Content (W/W): < 1.4%
Synonym: 2-Methylbutane

4. First-Aid Measures

Description of first aid measures

General advice:
No special precautions necessary.

If inhaled:
Keep patient calm, remove to fresh air. If difficulties occur: Seek medical attention.

If on skin:
Wash affected areas thoroughly with soap and water. If irritation develops, seek medical attention.

If in eyes:
In case of contact with the eyes, rinse immediately for at least 15 minutes with plenty of water. If irritation develops, seek medical attention.

If swallowed:
No hazards anticipated. Rinse mouth and then drink 200-300 ml of water. If difficulties occur: Seek medical attention.

Most important symptoms and effects, both acute and delayed

Symptoms: headache, dizziness, coordination disorder, dazed state, Eye irritation, skin irritation
Hazards: No hazards anticipated.

Indication of any immediate medical attention and special treatment needed

Note to physician
Treatment: Treat according to symptoms (decontamination, vital functions), no known specific antidote.

5. Fire-Fighting Measures

Extinguishing media

Suitable extinguishing media:
water spray, foam, dry powder, carbon dioxide

Unsuitable extinguishing media for safety reasons:
water jet

Additional information:
Use extinguishing measures to suit surroundings.

Special hazards arising from the substance or mixture

Hazards during fire-fighting:
carbon monoxide, carbon dioxide, Styrene, aliphatic hydrocarbons
The substances/groups of substances mentioned can be released in case of fire.

Advice for fire-fighters

Protective equipment for fire-fighting:
Wear self-contained breathing apparatus and chemical-protective clothing.

Further information:
Product will burn on contact with flame or exposure to high temperature. Burns with dense emission of soot. Containers/tanks should be cooled with water spray. Dispose of fire debris and contaminated extinguishing water in accordance with official regulations. Flammable concentrations of propellant may accumulate on storage in closed containers.

6. Accidental release measures

Further accidental release measures:
High risk of slipping due to leakage/spillage of product. Shut off or stop source of leak.
Substance/product can form explosive mixture with air.

Personal precautions, protective equipment and emergency procedures

Sources of ignition should be kept well clear. Ensure adequate ventilation. Note that this gas is heavier than air and can spread along the ground in the direction of the wind. Beware of pits and confined spaces. Use antistatic tools. Vapours are heavy and collect in low areas. Avoid all sources of ignition: heat, sparks, open flame.

Environmental precautions

Do not allow to enter drains or waterways. Discharge into the environment must be avoided.

Methods and material for containment and cleaning up

For small amounts: Sweep/shovel up. Pack in tightly closed containers for disposal.
For large amounts: Pick up with vacuum equipment approved for use in hazardous locations. Pack in tightly closed containers for disposal.

Ensure adequate ventilation. Dispose of absorbed material in accordance with regulations. Avoid raising dust.

7. Handling and Storage

Precautions for safe handling

Handle in accordance with good industrial hygiene and safety practice. Upon delivery, trailer and/or container should be opened and allowed to vent for a minimum of one (1) hour before unloading. The substance/product in bead or expanded form generates static charges during handling which are difficult to dissipate due to the insulating properties. Take precautionary measures against static discharges. Use antistatic tools. Provide good room ventilation even at ground level (vapours are heavier than air). Containers should be opened carefully in well-ventilated areas to avoid static discharge. Maintain air circulation and ventilation at a minimum rate of six air changes per hour to prevent the formation of flammable concentrations. Ensure adequate ventilation. The substance/ product may be handled only by appropriately trained personnel.

Protection against fire and explosion:
The product is combustible. Vapours may form ignitable mixture with air. Keep away from heat. Prevent electrostatic charge - sources of ignition should be kept well clear - fire extinguishers should be kept handy. Avoid all sources of ignition: heat, sparks, open flame. Containers should be earthed during decanting operations. It is recommended that all conductive parts of the machinery are grounded. All parts of the plant and equipment should be electrically bonded together and grounded. Electrical continuity should be checked at regular intervals. Higher line velocity can increase the build-up of static electric charge. Avoid flammable gas mixtures. Ensure an efficient ventilation (at least one air change per hour). Vapours are heavier than air and may accumulate in low areas and travel a considerable distance up to the source of ignition. Because of danger of explosion, avoid vapours reaching the cellar, sewage water and pits. Empty containers may contain flammable residue.

Conditions for safe storage, including any incompatibilities

Segregate from strong oxidizing agents.

Suitable materials for containers: High density polyethylene (HDPE), Low density polyethylene (LDPE), Paper/Fibreboard, Stove-lacquer RDL 50, Stove-lacquer R 78433, Carbon steel (Iron), Stainless steel 1.4301 (V2), Stainless steel 1.4361, Stainless steel 1.4401, Stainless steel 1.4439, Stainless steel 1.4539, Stainless steel 1.4541, Stainless steel 1.4571, Stainless steel 1.4306 (V2A), Stainless steel 1.4307, Stainless steel 1.4311, Stainless steel 1.4404, Polyamide (PA)

Further information on storage conditions: Protect against heat. Keep away from sources of ignition - No smoking. Keep only in the original container. Keep container tightly sealed. Protect against moisture. Avoid direct sunlight. Protect containers from physical damage. The authority permits and storage regulations must be observed. Store protected against freezing. Keep tanks under inert gas. Air monitoring should be used to alert any build up of explosive mixtures. Equipment to be installed in an environment with potentially explosive atmospheres should conform to the requirements of ATEX Directive 94/9/EC. Ventilate freight container with open door for 30 minutes before unloading.

Storage stability:
Keep container tightly closed and dry.
Keep only in the original container in a cool, dry, well-ventilated place away from ignition sources, heat or flame.

8. Exposure Controls/Personal Protection

Components with occupational exposure limits

isopentane	ACGIH, US:	TWA value 1,000 ppm ;
Pentane	ACGIH, US:	TWA value 1,000 ppm ;
	OSHA Z1:	PEL 1,000 ppm 2,950 mg/m3 ;

Advice on system design:
Provide local exhaust ventilation to control vapours/mists.

Personal protective equipment

Respiratory protection:
Respiratory protection may not be required under normal operating conditions if adequate ventilation is provided. Wear respiratory protection if ventilation is inadequate. Wear a NIOSH-certified (or equivalent) organic vapour/particulate respirator.

Hand protection:
Chemical resistant protective gloves, non-static gloves (e.g. of leather)

Eye protection:
Safety glasses with side-shields. Tightly fitting safety goggles (chemical goggles).

Body protection:
Anti-static protective clothing, Antistatic safety shoes

General safety and hygiene measures:
Avoid inhalation of dusts/mists/vapours. When using do not eat or drink. When using do not smoke.

9. Physical and Chemical Properties

Form:	beads	
Odour:	almost odourless	
Odour threshold:	not determined	
Colour:	white	
pH value:	not soluble	
softening temperature:	approx. 160 °F	
Freezing point:	No data available.	
onset of boiling:	The substance / product decomposes therefore not determined.	
Sublimation point:	not applicable	
Flash point:	Vapours are flammable.	
Flammability:	not highly flammable	(UN Test N.1 (ready combustible solids))
Flammability of Aerosol Products:	not applicable, the product does not form flammable aerosoles	
Lower explosion limit:	Product not examined: Value is calculated from the data of the components.	
Upper explosion limit:	Product not examined: Value is calculated from the data of the components.	
Autoignition:	285 °C	(DIN 51794)

Vapour pressure:	not applicable
Density:	approx. 1.02 - 1.05 g/cm3 (20 °C)
Relative density:	1.02 - 1.05 (20 °C)
Bulk density:	approx. 600 kg/m3 (20 °C)
Vapour density:	2.5 Heavier than air.
Partitioning coefficient n-octanol/water (log Pow):	not applicable
Self-ignition temperature:	not self-igniting
Thermal decomposition:	approx. 220 °C No decomposition if used as directed.
Viscosity, dynamic:	not applicable, the product is a solid
Viscosity, kinematic:	not applicable, the product is a solid
Particle size:	No data available.
Solubility in water:	not soluble not soluble
Miscibility with water:	immiscible
Solubility (quantitative):	No applicable information available.
Solubility (qualitative):	soluble solvent(s): aromatic hydrocarbons, ketones, organic solvents,
Evaporation rate:	The product is a non-volatile solid.

10. Stability and Reactivity

Reactivity
No hazardous reactions if stored and handled as prescribed/indicated., Vapours may form explosive mixture with air.

Corrosion to metals:
No corrosive effect on metal.

Oxidizing properties:
not fire-propagating

Chemical stability
The product is stable if stored and handled as prescribed/indicated.

Possibility of hazardous reactions
Formation of explosive gas/air mixtures.

Conditions to avoid
> 70 degrees Celsius
Avoid all sources of ignition: heat, sparks, open flame. Avoid direct sunlight. Avoid electro-static discharge.

Incompatible materials
explosive substances according UN transport regulations class 1, Propellant release will be boosted with increasing temperature.

Hazardous decomposition products

Decomposition products:
Possible thermal decomposition products: Pentane, styrene monomers, Heated product evolves combustible vapours.

Thermal decomposition:
approx. 220 °C
No decomposition if used as directed.

11. Toxicological information

Primary routes of exposure

Routes of entry for solids and liquids are ingestion and inhalation, but may include eye or skin contact. Routes of entry for gases include inhalation and eye contact. Skin contact may be a route of entry for liquefied gases.

Acute Toxicity/Effects

Acute toxicity
Assessment of acute toxicity: Contact with heated product can cause thermal burns.

Information on: isopentane
Assessment of acute toxicity:Virtually nontoxic after a single ingestion. Virtually nontoxic by inhalation. The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

Oral
Type of value: LD50
Value: > 2,000 mg/kg

Inhalation
Type of value: LC50
Value: > 5 mg/l

Dermal
Type of value: LD50
Value: > 2,000 mg/kg

Irritation / corrosion
Assessment of irritating effects: Frequent and prolonged contact can lead to skin irritation.
Irritation is possible when the product comes in contact with the skin, respiratory tract or the eyes.

Skin
Prolonged contact with the product can result in skin irritation.

Eye
Similar findings as for skin apply to eyes.

Sensitization
Assessment of sensitization: There is no evidence of a skin-sensitizing potential.

Result: Non-sensitizing.

Chronic Toxicity/Effects

Repeated dose toxicity
Assessment of repeated dose toxicity: Based on our experience and the information available, no adverse health effects are expected if handled as recommended with suitable precautions for designated uses.

Information on: isopentane
Assessment of repeated dose toxicity: Chronic overexposure has been shown to cause adverse kidney effects in experimental animals.

Genetic toxicity
Assessment of mutagenicity: Based on our experience and the information available, no adverse health effects are expected if handled as recommended with suitable precautions for designated uses.

Carcinogenicity
Assessment of carcinogenicity: Based on our experience and the information available, no adverse health effects are expected if handled as recommended with suitable precautions for designated uses.

Reproductive toxicity
Assessment of reproduction toxicity: Based on our experience and the information available, no adverse health effects are expected if handled as recommended with suitable precautions for designated uses.

Other Information
No reports of ill effects provided product was correctly handled and processed.

Information on: Pentane
Has a degreasing effect on skin.
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12. Ecological Information

Toxicity

Aquatic toxicity
Assessment of aquatic toxicity:
There is a high probability that the product is not acutely harmful to aquatic organisms. No toxic effects occur within the range of solubility.

Aquatic invertebrates
EC50 (48 h) > 100 mg/l, Daphnia magna (OECD Guideline 202, part 1, static)
Nominal concentration. The product has low solubility in the test medium. An eluate has been tested. No toxic effects occur within the range of solubility. The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

Aquatic plants
EC50 (72 h) > 100 mg/l (growth rate), Desmodesmus subspicatus (OECD Guideline 201, static)
Nominal concentration. The product has low solubility in the test medium. An eluate has been tested. No toxic effects occur within the range of solubility. The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

Aquatic toxicity

Information on: Pentane
Assessment of aquatic toxicity:
Acutely toxic for aquatic organisms. Toxic to aquatic life with long lasting effects.
EU-classification

Persistence and degradability

Assessment biodegradation and elimination (H2O)
In accordance with the required stability the product is not readily biodegradable. The product has not been tested. The statement has been derived from the structure of the product. The product is virtually insoluble in water and can thus be separated from water mechanically in suitable effluent treatment plants.

On the basis of the data available concerning eliminability/degradation and bioaccumulation potential, longer-term harm to the environment is improbable. No data available concerning biodegradation and elimination.

Elimination information

Non-biodegradable.

Bioaccumulative potential

Bioaccumulation potential
The product will not be readily bioavailable due to its consistency and insolubility in water.

Mobility in soil

Assessment transport between environmental compartments
Study scientifically not justified.

Additional information

Add. remarks environm. fate & pathway:
Because of the product's consistency and low water solubility, bioavailability is improbable.

Other ecotoxicological advice:
At the present state of knowledge, no negative ecological effects are expected. No toxic effects occur within the range of solubility.

Information on: Pentane
Other ecotoxicological advice:
The substance has a very low Global Warming Potential and no Ozone Depleting Potential.
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13. Disposal considerations

Waste disposal of substance:
This product is not regulated by RCRA. This product is not regulated by CERCLA ('Superfund'). Dispose of in accordance with national, state and local regulations. Do not discharge into waterways or sewer systems without proper authorization.

Container disposal:
Remove all packaging for recovery or disposal

14. Transport Information

Land transport
USDOT

Hazard class:	9
Packing group:	III
ID number:	UN 2211
Hazard label:	9
Proper shipping name:	POLYMERIC BEADS, EXPANDABLE

Sea transport
IMDG

Hazard class:	9
Packing group:	III
ID number:	UN 2211
Hazard label:	9
Marine pollutant:	NO
Proper shipping name:	POLYMERIC BEADS, EXPANDABLE

Air transport
IATA/ICAO

Hazard class:	9
Packing group:	III
ID number:	UN 2211
Hazard label:	9
Proper shipping name:	POLYMERIC BEADS, EXPANDABLE

15. Regulatory Information

Federal Regulations

Registration status:
Chemical TSCA, US

All substances are TSCA listed and active.

EPCRA 311/312 (Hazard categories): Refer to SDS section 2 for GHS hazard classes applicable for this product.

CERCLA RQ	CAS Number	Chemical name
10 LBS	78-78-4; 109-66-0	isopentane; Pentane

Safe Drinking Water & Toxic Enforcement Act, CA Prop. 65:

WARNING: This product can expose you to chemicals including BENZENE, which is known to the State of California to cause cancer and birth defects or other reproductive harm. For more information, go to www.P65Warnings.ca.gov.

NFPA Hazard codes:
Health: 1 Fire: 2 Reactivity: 0 Special:

HMIS III rating
Health: 1 Flammability: 2 Physical hazard: 0

Assessment of the hazard classes according to UN GHS criteria (most recent version):

16. Other Information

SDS Prepared by:
BASF NA Product Regulations
SDS Prepared on: 2025/02/10

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